

Theoretical Analyses of Epidemiological Models

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Epidemics on Networks

Given the ongoing COVID-19 pandemic, epidemiological models have become critical to study for public health and economy.

Most mathematical models assume that set of contacts of each individual is random and changing. Networks, however, enable us to capture the permanence of interactions. Moreover, they give information about the individual-level behaviour of infection, epidemic patterns, and the design of intervention strategies to combat epidemics.

SIR Model on Graphs

SIR Model

Most mathematical models uses compartmentalization according to individuals' disease status. In the case of the **SIR Model**,



Figure: SIR Model with different rates for movement between compartments

the force of infection depends on infected proportion of the population and **transmission rate** β . The **recovery rate** ρ is based on the rate at which individuals recover.

Network Models

These models take a network of individuals, called *nodes*. Two nodes are connected by an edge if they can transmit the disease to each other.

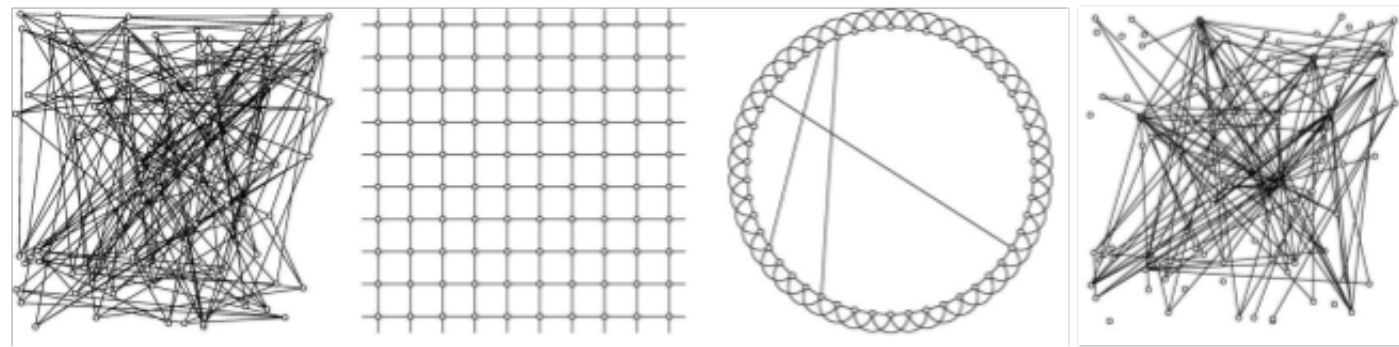


Figure: Image from Keeling et al, 2005: From left to right: random graph, lattice, small world, and scale-free (power-law model)

SIR on Graph

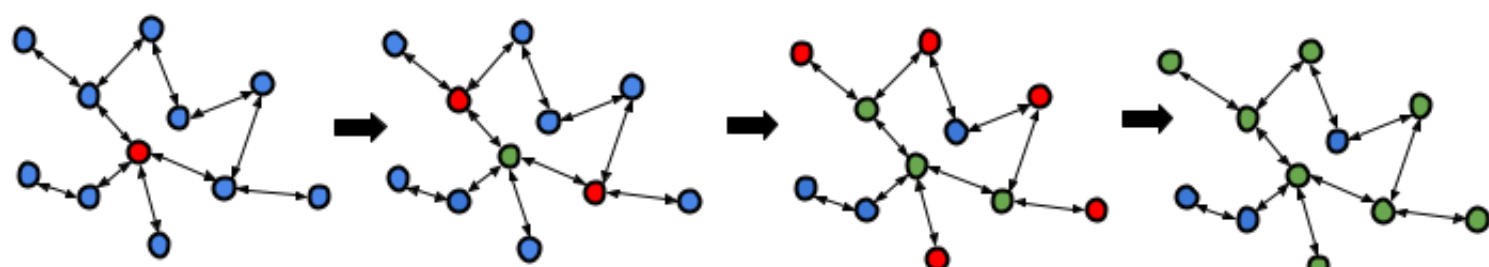


Figure: Spread of an epidemic on a Networked SIR Model

The rates of movements between S, I and R can be formalized mathematically using the adjacency matrix and differential equations.



Figure: Star Graph with exponential random variables

Let p be Pr [a directed edge transmits a disease before the node recovers] in this model where β is transmission rate and ρ is recovery rate. Then,

$$p = \int_0^\infty \beta e^{\beta x} \int_x^\infty \rho e^{\rho y} dy dx = \frac{\beta}{\beta + \rho}$$

Main Question

What structural properties of networks affect the spread of an epidemiological disease in different models?

Percolation Model

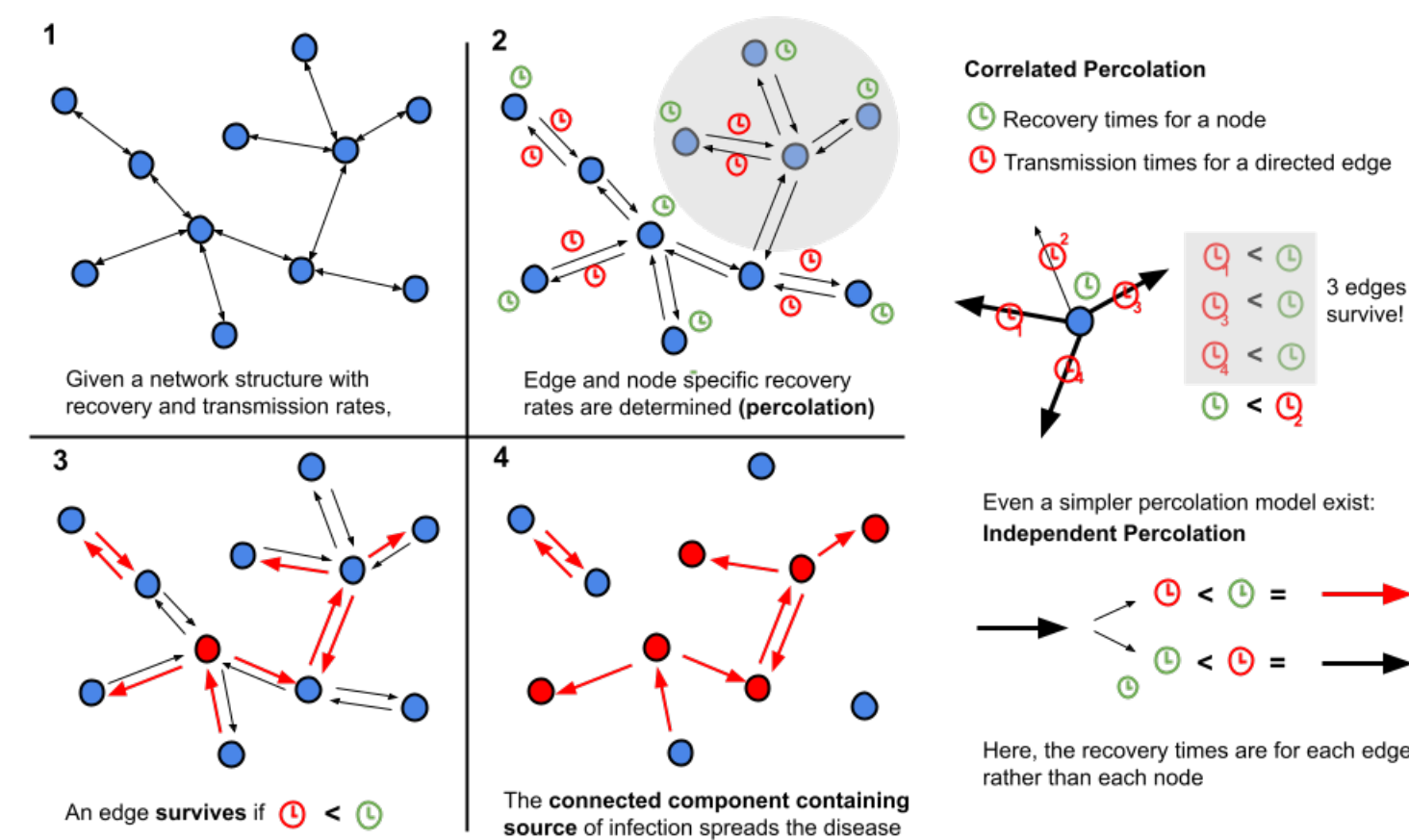


Figure: Percolation Process

In Percolation Model, each edge survives with *percolation probability* $p = \frac{\beta}{\beta + \rho}$.

How many nodes get infected?

As *percolation probability* p increases, at a certain threshold, a **phase transition** is observed in the size of the infection.

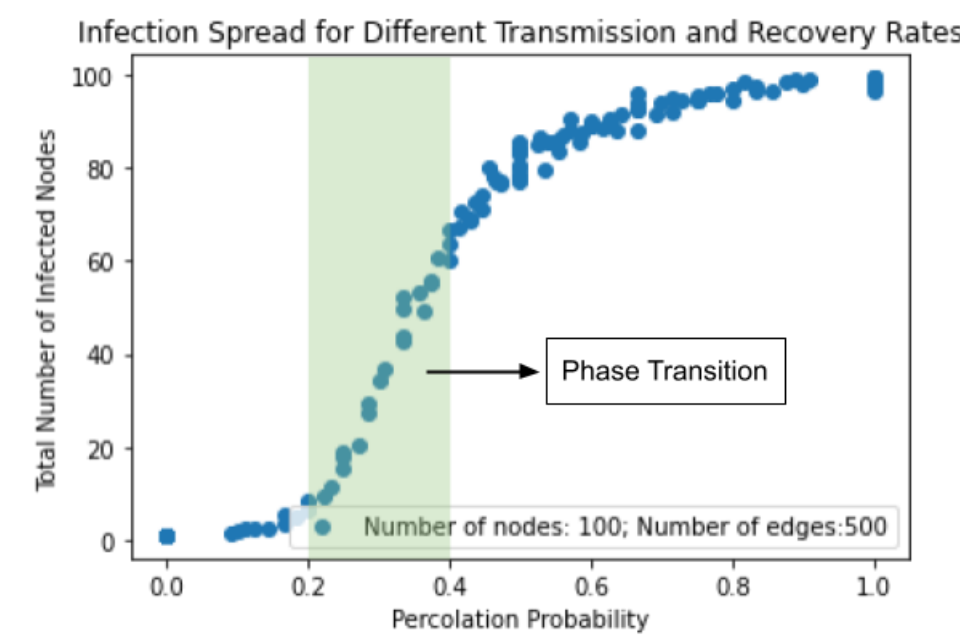


Figure: Phase Transition in Erdos Renyi Graph with 100 nodes and 500 edges

When the *expected degree of a node* exceeds **1**, a **giant component** consisting of a constant fraction of the nodes emerges. It contains the Strongly Connected Component which spreads the epidemic throughout itself!

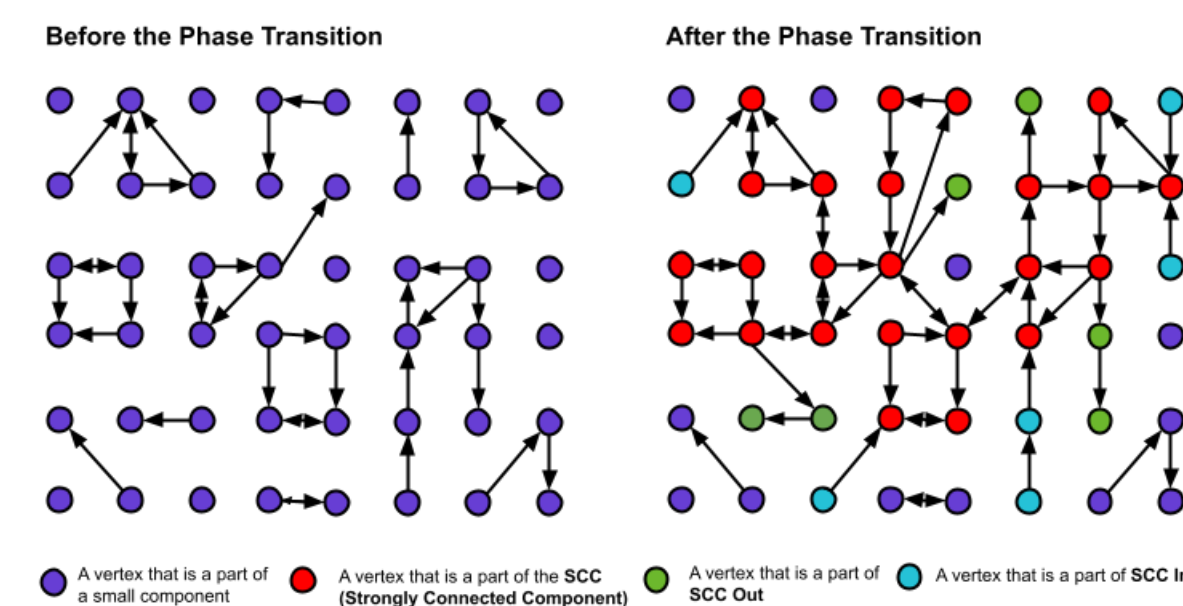


Figure: SCC structure before and after Phase Transition

Calculating the Size of the Giant Component

For some graph such as random graphs with a given node degree distribution, we can calculate the half-edge **degree distribution**.

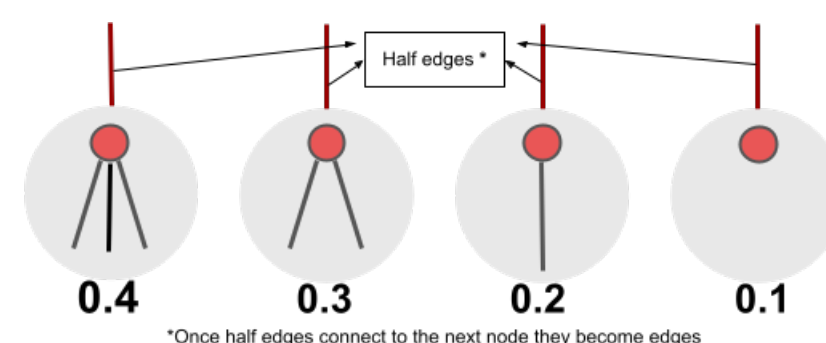


Figure: Edge Degree Distribution of a graph where the probability that a node has a degree of 1, 2, 3, and 4 is 0.25 respectively.

A **branching process** is a method for creating a random tree starting with the root half edge and following the edge degree distribution at each generation.

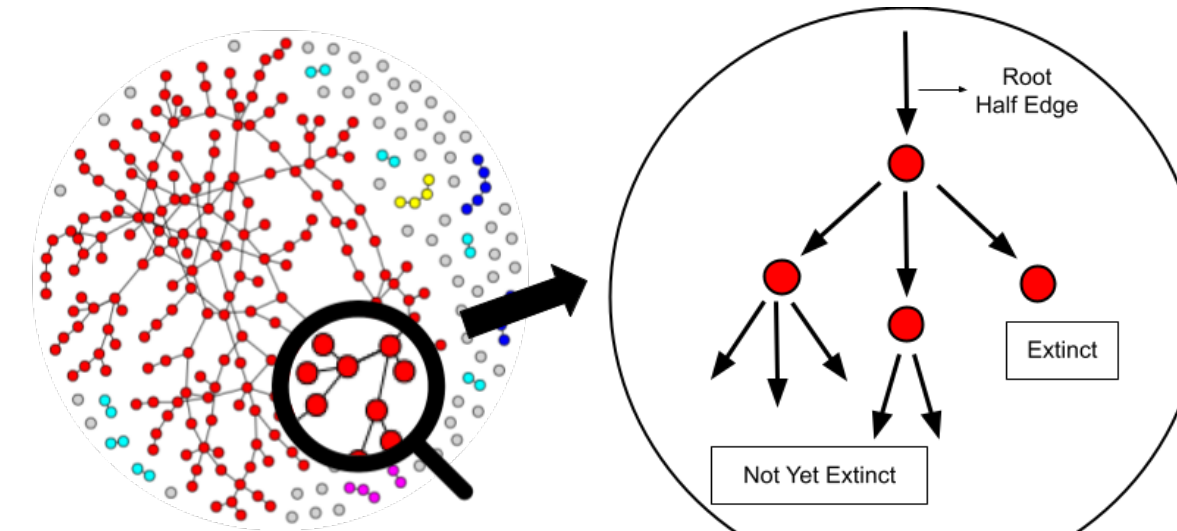


Figure: Image from Norris and Darling, 2010: Locally tree-like Erdos Renyi Random Graph

Using node and half-edge degree distributions, we can calculate the *extinction probability* X , the probability that a half edge dies out.

Extinction Probability:

$$X = \sum_{k=1}^{\infty} Pr[\text{a half edge has } k \text{ children}] X^k$$

Then, if at least one of the half-edges stemming from a node does not die out, then that node is in the Giant Component.

Size of the Giant Component: Let $P_k = Pr$ [a node has degree k] and n be number of nodes.

$$E[\text{Size of the Giant Component}] = n \sum_{k=0}^{\infty} P_k (1 - X^k)$$

Correlated vs Independent Percolation

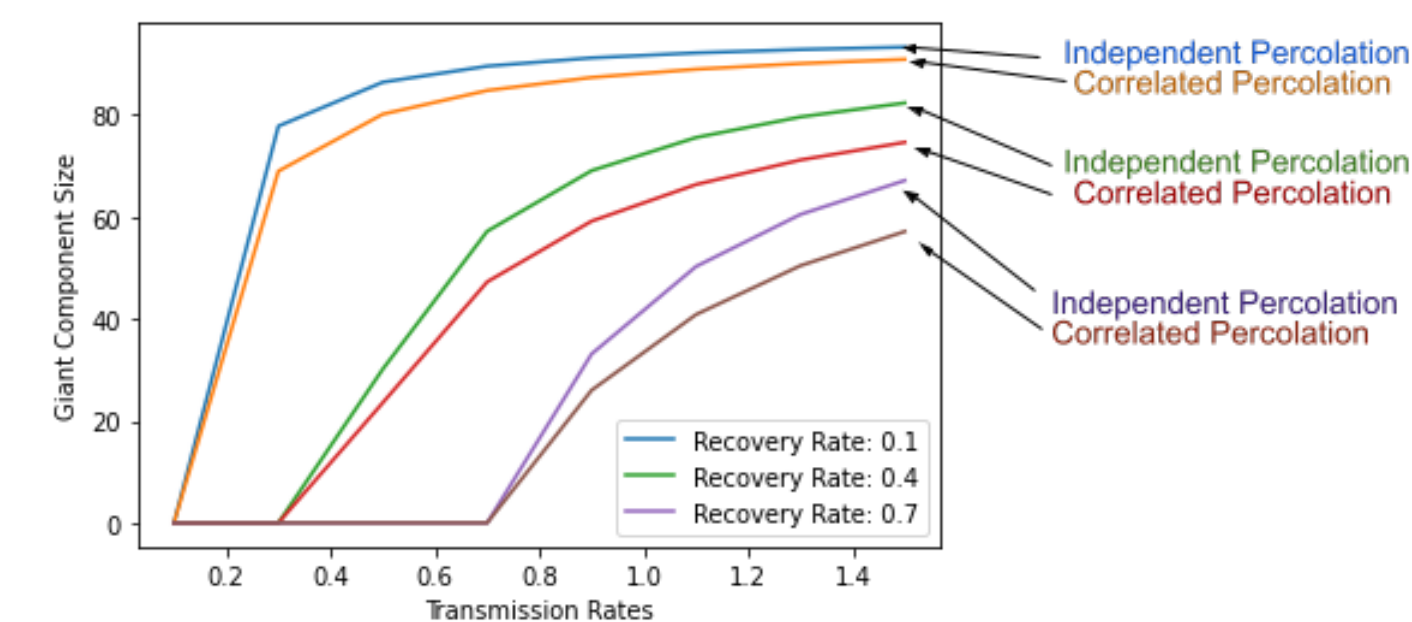


Figure: Size of the Giant Components after Percolation of a Graph where the probability that each node has degree 1, 2, 3, 4 is 0.25 respectively

Observation (1):

Epidemic threshold for both independent and correlated percolation is the same.

Observation (2): Conjecture**

Independent percolation is an **upper bound** for correlated percolation.

**Possible proof using Holley's Condition and FKG Inequality

But does (1) always hold?

A Nested Triangular Configuration Model:

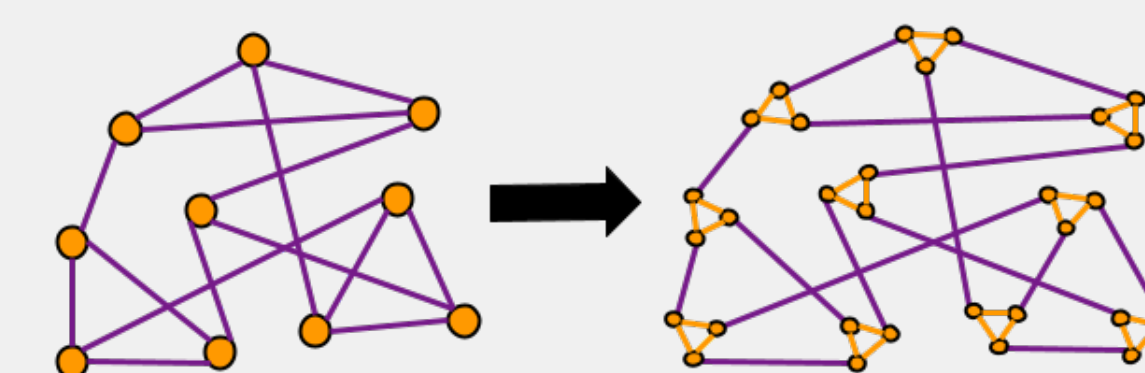


Figure: 3-regular graph obtained by replacing each node of an undirected 3-regular graph with a triangle

This graph models a type of **Household Structure**, which is a way to structure communities to combat epidemics.

Why does this violate (1)?

Removing certain edges affects the spread of the epidemic within the triangle households (**site percolation**) differently in two percolation models in addition to nodes connected to removed edge (**bond percolation**).

Analysis: Colorful Branching Process

Since branching process does not self-loop, small cycles within the triangular configuration model made it hard to use branching process. We colored the edges and grouped half-edges into 4 types based on what color they and their parent edges are.

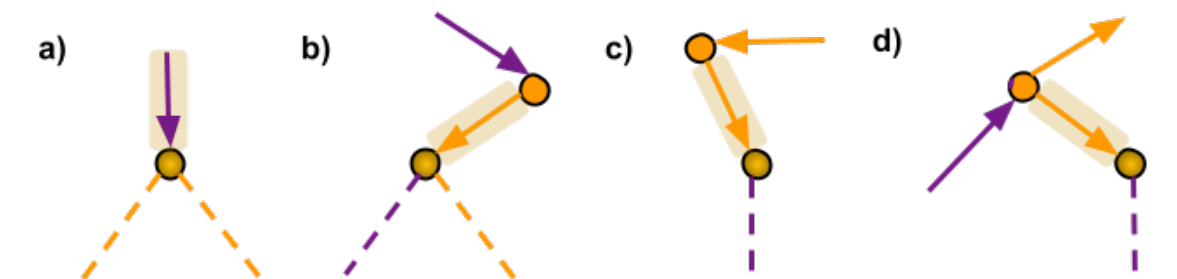


Figure: Half-edge (a), (b), (c), and (d) highlighted above have extinction probabilities $X_{(a)}$, $X_{(b)}$, $X_{(c)}$ and $X_{(d)}$ respectively

Based on the type of their half-edge children, the *extinction probability* of each type is calculated. Then, based on the *colored* degree distribution, the size of the giant component after each type of percolation is computed.

A Counterexample to (1)

The *phase transition* for Correlated Percolation occurs at a **higher epidemic threshold** than the one for (Independent) Percolation!

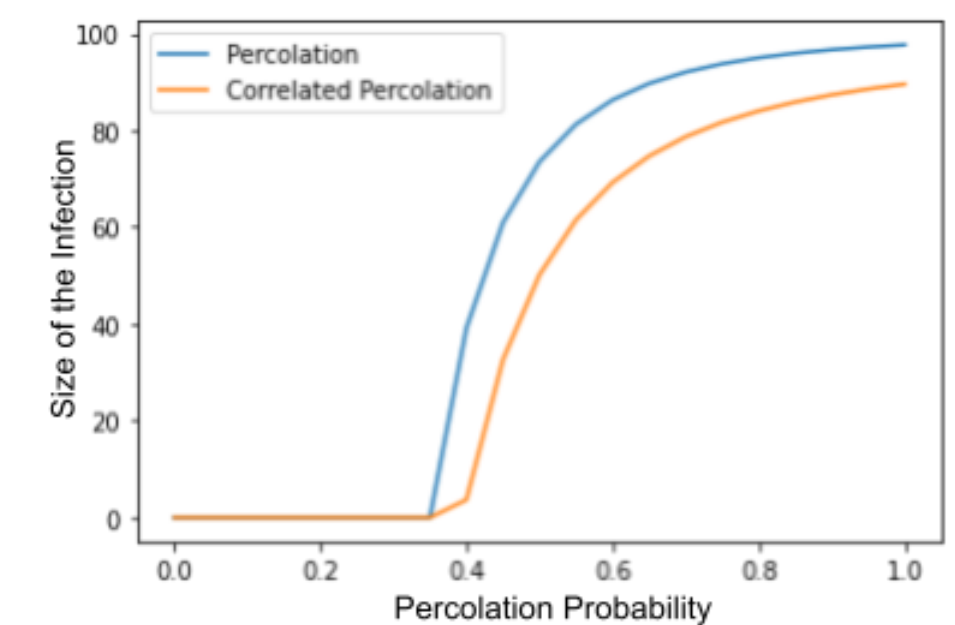


Figure: Size of the Giant Component after Independent and Correlated Percolation

Conclusion

The spread of the epidemic within the model and the extent to which different models substitute one another depends heavily on what kind of graph is chosen to represent the social network of individuals.

Next Steps: How to immunize the households?

Based on our experiments the best strategy to immunize nodes is *uniform distribution*.

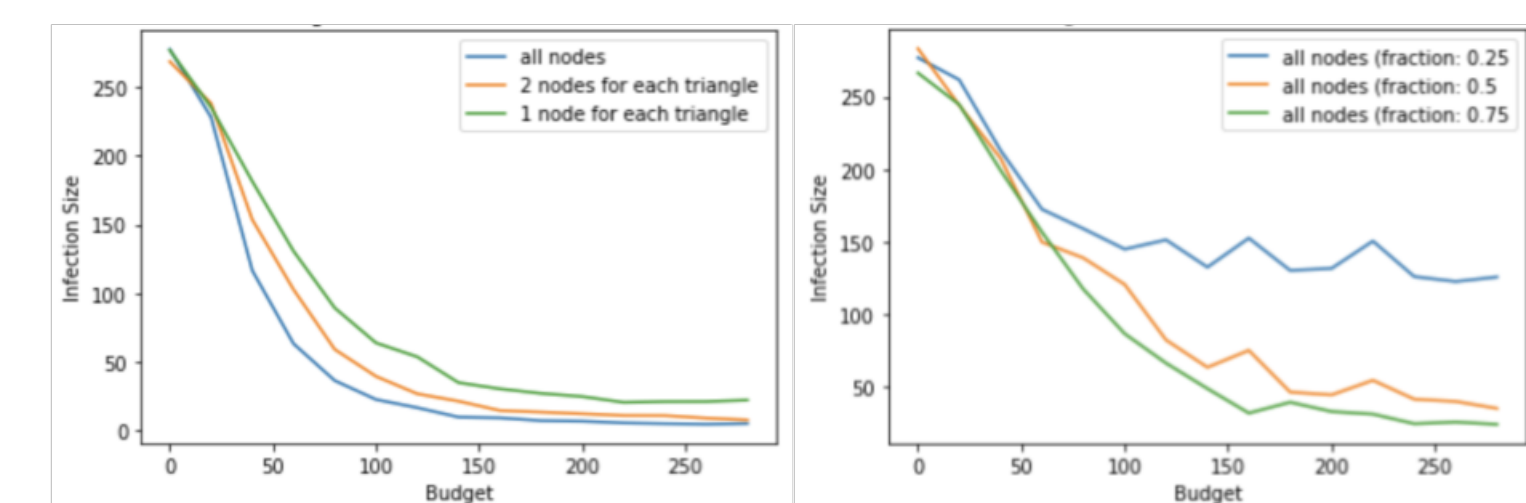


Figure: Targeting different fraction of triangles and nodes within triangles

Further Research Areas: There exists many more strategies to combat epidemics on different graphs with different degree distributions and amounts of clustering. A variety of tools such as network eigenvalues, spectral partitioning, convex optimization, recursive algorithms, effective resistances, adjacency matrix, new frameworks based on heuristic, etc are used.